Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-E Primary Myocardial Diseases

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-05-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild dilated cardiomyopathy from its central defect

A competent response contains

· The central defect is loss of contractile force: myocytes generate less tension, so stroke volume falls and end-systolic volume rises.

· Compensation runs through the Frank-Starling mechanism and neurohormonal activation, with salt and water retention raising end-diastolic volume to restore some stroke volume.

· Sarcomeres are added in series, which is eccentric hypertrophy, so myocytes lengthen and the cavity enlarges while the wall stays normal or thin even though total heart weight rises.

· Law of Laplace: wall stress is proportional to pressure times radius divided by wall thickness.

· A wider, thinner ventricle must develop more wall tension for the same pressure, which raises oxygen demand, injures more myocytes, and drives further dilation, so the ejection fraction falls, usually below about 40%.

· Familial disease accounts for about 30% to 50% of cases, usually autosomal dominant, most often truncating variants of TTN (titin), also lamin A/C and X-linked dystrophin.

· Alcohol: ethanol and acetaldehyde damage mitochondria and contractile proteins, and the picture may improve with abstinence.

· Prior myocarditis: myocyte loss and fibrosis after viral or immune injury.

· Doxorubicin: dose-dependent, through iron-dependent free radical injury and damage to topoisomerase II beta in myocytes.

· Peripartum: late pregnancy to about 5 months postpartum, with oxidative stress activating cathepsin D, which cleaves prolactin to an antiangiogenic 16 kDa fragment.

· Chagas disease: chronic Trypanosoma cruzi infection with inflammation and fibrosis, often an apical aneurysm.

· Thiamine deficiency (wet beriberi): loss of thiamine pyrophosphate, the cofactor for pyruvate and alpha-ketoglutarate dehydrogenase, lowers ATP, and vasodilation adds high-output failure.

· Hemochromatosis: iron catalyzes free radical (Fenton) injury, giving a dilated pattern and sometimes a restrictive one.

Distinctions that matter

· Eccentric hypertrophy adds sarcomeres in series and enlarges the cavity; concentric hypertrophy adds them in parallel and thickens the wall at the expense of the cavity.

· Heart weight and wall thickness move in different directions here: the heart is heavier while the wall is normal or thin, because the muscle is spread around a much larger cavity.

Common misconceptions

· Writing that the wall thickens in dilated disease. Growth is in series, so the chamber widens rather than the wall thickening.

· Treating dilation as purely compensatory. Larger radius with a thin wall raises wall stress, which is what makes the process self-perpetuating.

Where it is taught: Section C01, first two paragraphs and the causes table of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C01-BD2

Brain dump 2 · integration and application: Explain every finding in a large, weak ventricle

A competent response contains

· Both murmurs are functional regurgitation: dilation stretches the mitral and tricuspid annuli, pulling the leaflets apart so they fail to coapt, giving holosystolic murmurs with structurally normal valves. The apical murmur is mitral and the one at the left lower sternal border is tricuspid.

· The early diastolic extra sound is an S3, produced when blood rushing into an overfilled ventricle in early diastole is abruptly decelerated.

· Low output explains fatigue, and high filling pressures explain dyspnea and orthopnea.

· Raised jugular venous pressure and peripheral edema come from high right-sided filling pressures and volume retention.

· Stasis in dilated, poorly contracting chambers favors mural thrombus and systemic or pulmonary emboli.

· Stretched, fibrotic myocardium supports atrial fibrillation and ventricular tachycardia.

· Chest radiography shows a globular, enlarged heart.

· Ischemic cardiomyopathy produces the same dilated, low ejection fraction ventricle from prior infarcts and hibernating myocardium, but the wall motion loss is regional rather than diffuse, and the history carries infarction or coronary disease.

· Histology in idiopathic or familial disease is nonspecific: hypertrophied myocytes with large nuclei, attenuated fibers, and interstitial fibrosis.

Distinctions that matter

· Functional regurgitation comes from annular geometry, not from leaflet or chordal disease, so the valve itself looks normal.

· Diffuse against regional wall motion loss is the feature that separates primary dilated disease from ischemic cardiomyopathy.

Common misconceptions

· Reading a holosystolic apical murmur in a dilated heart as primary mitral valve disease. The leaflets are normal and the annulus has been stretched.

· Attributing the S3 to valve closure. It is the sound of rapid early filling being abruptly decelerated by an overfilled ventricle.

Where it is taught: Section C01, paragraphs on findings and on distinguishing ischemic cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C01-RT01

Retrieval exercise: Rebuild the remodeling spiral

A competent response contains

· Step 1: myocyte injury or a defective sarcomere protein.

· Step 2: contractility and stroke volume fall.

· Step 3: volume retention and series sarcomeres enlarge the cavity.

· Step 4: a larger radius and thin wall raise wall stress, by the law of Laplace.

· Step 5: higher oxygen demand injures more myocytes, which returns the loop to step 1.

· The two processes that enlarge the cavity are neurohormonally driven salt and water retention, which raises end-diastolic volume, and eccentric hypertrophy adding sarcomeres in series.

· Wall stress is proportional to pressure times radius divided by wall thickness: radius is rising and wall thickness is falling, so both moving variables push wall stress up.

· Higher wall tension costs oxygen because developing and maintaining tension is what consumes myocardial ATP, so demand rises in muscle whose supply has not changed.

· Each turn of the loop enlarges the ventricle further and lowers the ejection fraction further.

Distinctions that matter

· Stroke volume can be partly restored by a larger end-diastolic volume while the ejection fraction still falls, because the denominator grows faster than the numerator.

· The loop is driven by geometry as well as by the original injury, which is why it continues after the initial insult has passed.

Common misconceptions

· Drawing the spiral as a straight line that ends in dilation. It is a closed loop, and that is why the disease progresses.

· Leaving wall thickness out of the Laplace step. A thinner wall raises wall stress as much as a wider radius does.

Where it is taught: Section C01, the remodeling spiral cycle of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-BD1

Brain dump 1 · foundational retrieval: Rebuild hypertrophic cardiomyopathy from gene to tissue to filling

A competent response contains

· Autosomal dominant with variable penetrance, affecting about 1 in 500 people, and the most common inherited heart disease.

· The genes most often involved are MYH7, encoding beta-myosin heavy chain, and MYBPC3, encoding myosin-binding protein C.

· Mutant sarcomeres are hypercontractile and use energy inefficiently, and the myocardium hypertrophies with no pressure overload on it.

· Hypertrophy is usually asymmetric and thickest in the interventricular septum, and the cavity is small.

· Histology: myocyte disarray, interstitial and replacement fibrosis, and thickened intramural coronary arteries.

· The main defect is diastolic: a thick, fibrotic wall is stiff and relaxes slowly, so filling needs high pressure and depends on atrial contraction.

· Atrial contraction into a stiff ventricle produces an S4, and atrial fibrillation is poorly tolerated because the atrial contribution to filling is lost.

· Systolic function is preserved, so the ejection fraction is normal or high.

· Exertional angina occurs with normal epicardial coronaries because thick muscle with narrowed intramural arteries outgrows its blood supply.

Distinctions that matter

· Hypertrophy here is independent of load, unlike the concentric hypertrophy of pressure overload, which is why no hypertension or valve lesion is needed to explain it.

· A normal or high ejection fraction coexists with heart failure, because the failure is one of filling rather than emptying.

Common misconceptions

· Reading a high ejection fraction as a healthy ventricle. The ejection fraction is a fraction of a small end-diastolic volume, and filling is the problem.

· Explaining the angina by coronary atherosclerosis. The epicardial arteries are normal and the mismatch comes from muscle mass and thickened intramural vessels.

Where it is taught: Section C02, first two paragraphs of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-BD2

Brain dump 2 · integration and application: Reason through obstruction, syncope, and sudden death

A competent response contains

· Sequence: septal hypertrophy narrows the left ventricular outflow tract; systolic flow accelerates through the narrow tract; drag pulls the anterior mitral leaflet toward the septum; leaflet contact obstructs outflow in mid systole; the malcoapting leaflet leaks, giving posteriorly directed mitral regurgitation.

· The rule: anything that reduces left ventricular volume (less preload or less afterload) or raises contractility brings leaflet and septum closer and worsens obstruction; anything that enlarges the ventricle separates them and relieves it.

· Standing from squatting drops venous return, so the cavity is smaller and the murmur is louder; squatting raises venous return and afterload, so the cavity is larger and the murmur is softer.

· The murmur is a harsh crescendo-decrescendo systolic murmur at the left lower sternal border that does not radiate well to the carotids.

· The carotid pulse rises briskly, then falls as obstruction develops, and may be bifid.

· Sudden death: myocyte disarray and patchy fibrosis create regions of slow, nonuniform conduction that support reentrant ventricular tachycardia and fibrillation, and this is the most commonly identified cause of sudden cardiac death in young athletes in the United States.

· Exertion adds catecholamines, a small cavity, and ischemia.

· Exertional syncope can come from ventricular arrhythmia or from obstruction that worsens as heart rate and contractility rise.

· The athlete's heart has symmetric physiologic hypertrophy, a normal or large cavity, normal diastolic filling, and regression with rest from training.

· The electrocardiogram often shows left ventricular hypertrophy voltage and deep, narrow septal Q waves.

Distinctions that matter

· The obstruction is dynamic rather than fixed, so it tracks cavity size and contractility beat to beat, which is why maneuvers change the murmur.

· Aortic stenosis responds the opposite way to Valsalva and standing, because a fixed orifice depends on the volume crossing it rather than on leaflet position.

Common misconceptions

· Saying anything that increases flow makes the murmur louder. Raising preload enlarges the cavity, moves the leaflet away from the septum, and softens it.

· Attributing the mitral regurgitation to primary valve disease. The leaflet is displaced toward the septum and therefore fails to coapt.

Where it is taught: Section C02, the obstruction flow, the maneuver table, the worked example, and the paragraph on sudden death of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-CP01

Comparison table: Hypertrophic cardiomyopathy against the athlete's heart

A competent response contains

· What drives the hypertrophy: sarcomere protein mutations that make sarcomeres hypercontractile and energy inefficient, independent of load, against physiologic adaptation to training.

· Symmetry and site: asymmetric, thickest in the interventricular septum, against symmetric thickening.

· Cavity size: small left ventricular cavity, against a normal or large cavity.

· Diastolic filling: impaired, with a stiff slowly relaxing wall and dependence on atrial contraction, against normal filling.

· After rest from training: hypertrophy persists in the disease, while the athlete's hypertrophy regresses.

· Tissue architecture: myocyte disarray with interstitial and replacement fibrosis and thickened intramural coronary arteries, against normal architecture.

· Risk of exertional arrhythmia and sudden death: high in the disease, because disarray and fibrosis create a reentry substrate, and not raised in the athlete's heart.

Distinctions that matter

· Cavity size and diastolic filling do most of the work in separating the two, because both conditions can show a thick wall.

· Regression with rest is a feature of adaptation, not of disease.

Common misconceptions

· Using wall thickness alone to decide. Thickness overlaps between the two, so symmetry, cavity size, and filling are what separate them.

· Assuming the athlete's thick wall also carries an arrhythmia risk. The architecture is normal, so the reentry substrate is absent.

Where it is taught: Section C02, paragraph on sudden death and the athlete's heart, with the opening paragraph on genetics and histology of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-PR01

Prediction exercise: Predict what the strain phase of Valsalva does in hypertrophic cardiomyopathy

A competent response contains

· Venous return to the right heart falls, because raised intrathoracic pressure during strain impedes systemic venous inflow.

· Left ventricular end-diastolic volume falls, because less blood reaches the left ventricle to fill it.

· The distance between the anterior mitral leaflet and the septum falls, because a smaller cavity brings the two closer together.

· Systolic anterior motion begins earlier and is greater, because the leaflet starts nearer the septum and drag from accelerated flow pulls it across sooner.

· The outflow murmur becomes louder, because the dynamic obstruction is greater.

· The murmur of aortic stenosis becomes softer, because a fixed orifice murmur depends on the volume crossing it and less blood is being ejected.

· Squatting and passive leg raising both raise venous return and enlarge the ventricle, separating leaflet from septum, so the murmur becomes softer; squatting also raises afterload, which enlarges the cavity further.

Distinctions that matter

· Valsalva and standing separate the two murmurs precisely because one obstruction is dynamic and the other is fixed.

· Afterload and preload both act through cavity size here, which is why handgrip and leg raising change the murmur in the same direction.

Common misconceptions

· Predicting that both murmurs change in the same direction with Valsalva. They move opposite ways, and that is the point of the maneuver.

· Thinking the leaflet moves because of the murmur. Flow acceleration and drag move the leaflet, and the murmur reports the resulting obstruction.

Where it is taught: Section C02, the obstruction flow, the maneuver table and its caption of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-BD1

Brain dump 1 · foundational retrieval: Rebuild restrictive cardiomyopathy and its causes

A competent response contains

· The wall stiffens either because something is deposited in it or because myocardium and endocardium are replaced by fibrous tissue.

· Compliance is the change in ventricular volume per change in filling pressure, so a stiff ventricle has low compliance and its filling curve is steep.

· A small increase in volume therefore produces a steep rise in diastolic pressure.

· Early diastolic filling is rapid and then stops abruptly when the rigid wall reaches its limit, which gives the dip-and-plateau or square root pattern on a ventricular pressure tracing.

· Contraction is relatively preserved, so cavity size and ejection fraction are near normal until late disease.

· Amyloidosis is the most common cause: extracellular AL light chain or transthyretin amyloid, with low electrocardiogram voltage despite thick walls, Congo red staining with apple-green birefringence, myeloma in AL disease, and older men in wild-type transthyretin disease.

· Sarcoidosis: noncaseating granulomas and scar, with heart block and ventricular arrhythmia, and lung and node disease.

· Hemochromatosis: intracellular iron and fibrosis, with diabetes, skin pigmentation, and liver disease, and it may be dilated instead.

· Radiation: interstitial and endocardial fibrosis after chest irradiation, and it may coexist with constriction.

· Endomyocardial fibrosis: fibrosis of the apex and inflow tracts in children and young adults in tropical Africa and Asia.

· Loeffler endocarditis: eosinophil-driven endocardial injury, then thrombus, then fibrosis, in hypereosinophilic syndrome with marked blood eosinophilia.

· The three stages in Loeffler endocarditis are eosinophil granule proteins such as major basic protein damaging the endocardium, mural thrombus forming on the injured surface, and organization into fibrosis that obliterates the apex.

· Restrictive cardiomyopathy is the least common of the three cardiomyopathies.

Distinctions that matter

· Infiltrative causes thicken the wall, while endomyocardial causes obliterate the cavity from the inside.

· Compliance and contractility are separate properties: compliance is lost here while contractility is largely intact.

Common misconceptions

· Expecting a low ejection fraction. Emptying is preserved and filling is what fails, so the ejection fraction stays near normal until late.

· Describing filling as uniformly slow. Early filling is rapid and then halts abruptly, which is what makes the dip-and-plateau shape.

Where it is taught: Section C03, first paragraph, the causes table, and the Loeffler paragraph of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-BD2

Brain dump 2 · integration and application: Trace high filling pressures to the bedside and read the ECG mismatch

A competent response contains

· High filling pressures are transmitted backward into the atria, so both atria dilate markedly.

· Atrial dilation predisposes to atrial fibrillation and to thrombus.

· Right-sided congestion often dominates: raised jugular venous pressure that may rise on inspiration, which is the Kussmaul sign, with hepatomegaly, ascites, and peripheral edema, along with dyspnea from pulmonary congestion.

· Cardiac output is fixed because stroke volume cannot be raised by filling the ventricle more, so exercise tolerance is poor.

· Low voltage sequence: amyloid fibrils deposit between myocytes; the echocardiogram shows a thick, often speckled wall; deposits add mass but no electrically active cells; myocytes atrophy and are separated by protein; less depolarizing muscle gives low QRS voltage.

· In true hypertrophy, as in hypertrophic cardiomyopathy or hypertension, added muscle raises voltage, so the same thickness would give tall complexes.

· The mismatch of low QRS voltage in the limb leads with increased wall thickness is the classic clue to cardiac amyloidosis, along with Q waves without infarction (pseudoinfarct pattern) and conduction disease.

· Endomyocardial biopsy shows pink extracellular material that stains with Congo red and shows apple-green birefringence under polarized light.

· Constrictive pericarditis produces a similar picture, but the restriction comes from the pericardium rather than the myocardium.

Distinctions that matter

· Measured wall thickness and electrical mass are different things, which is the whole basis of the amyloid clue.

· Restrictive cardiomyopathy and constrictive pericarditis share the physiology of failed filling and raised venous pressure, and they are separated by where the restriction sits.

Common misconceptions

· Reading a thick wall as evidence of hypertrophy. In amyloidosis the thickness is extracellular protein, not myocytes.

· Expecting stroke volume to rise with exercise through greater filling. The ventricle cannot accept more volume, so output is fixed.

Where it is taught: Section C03, second paragraph, the amyloid flow, and the paragraph on the voltage mismatch of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-BD1

Brain dump 1 · foundational retrieval: Sort the three cardiomyopathies by which half of the cycle fails

A competent response contains

· The first question is which half of the cardiac cycle fails, emptying or filling.

· Dilated disease: the ventricle cannot empty, contractility is lost, end-systolic volume rises, and the ejection fraction falls, which is systolic dysfunction and heart failure with reduced ejection fraction.

· Hypertrophic disease: the ventricle cannot fill, contraction is preserved, and the stiff wall requires high filling pressure, which is diastolic dysfunction and heart failure with preserved ejection fraction.

· Restrictive disease: the ventricle cannot fill for the same reason, with preserved contraction and a preserved ejection fraction.

· Hypertrophic disease is stiff because of thick, disorganized muscle; restrictive disease is stiff because of infiltration or fibrosis.

· Dilated loop: lower contractility flattens the end-systolic pressure-volume relationship, so the loop shifts right to larger volumes and narrows.

· Hypertrophic loop: the loop shifts left to smaller volumes and the end-diastolic pressure-volume relationship is steep.

· Restrictive loop: a steep end-diastolic pressure-volume relationship with a small loop.

· The end-systolic pressure-volume relationship represents contractility and the end-diastolic pressure-volume relationship represents compliance, so a flatter end-systolic line means weaker contraction and a

steeper end-diastolic curve means a stiffer ventricle in which every filling volume costs more pressure, and that pressure is what the lungs and systemic veins feel.

· All three share one consequence: scarred, stretched, or disorganized myocardium supports reentry, so ventricular arrhythmia and sudden death occur in each.

Distinctions that matter

· Contractility and compliance are read from different lines on the pressure-volume diagram, which is why systolic and diastolic failure look different there.

· Hypertrophic and restrictive disease share the physiology of failed filling and differ in what makes the wall stiff.

Common misconceptions

· Equating a preserved ejection fraction with a normal heart. Filling pressure is high in both diastolic diseases, which is what produces the symptoms.

· Drawing the dilated loop as larger in every dimension. It shifts to larger volumes and narrows, because stroke volume is reduced.

Where it is taught: Section C04, first paragraph, the comparison table, and the paragraph on pressure-volume reasoning of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-BD2

Brain dump 2 · integration and application: Classify four sets of measurements

A competent response contains

· Patient A: emptying has failed. An enlarged cavity, a 7 mm wall, and an ejection fraction of 20% give dilated cardiomyopathy, settled by the low ejection fraction with a large cavity. The loop shifts right and narrows with a flatter end-systolic line, and the early diastolic extra sound is an S3.

· Patient B: filling has failed. Asymmetric septal thickening of 21 mm against a 10 mm posterior wall, a small cavity, an ejection fraction of 72%, and the anterior mitral leaflet touching the septum in systole give hypertrophic cardiomyopathy with obstruction, settled by the asymmetric thickening with systolic anterior motion. The loop shifts left with a steep end-diastolic curve, and the expected extra sound is an S4.

· Patient C: filling has failed. A normal cavity, a 16 mm wall, an ejection fraction of 55%, huge atria, and low limb lead voltage give restrictive disease from amyloid, settled by the low voltage with a thick wall and enlarged atria. The loop is small with a steep end-diastolic curve, and an S3 from the abrupt halt to filling is expected, with an S4 if the atria still contract.

· Patient D: filling has failed. A 19 mm wall with tall QRS complexes, a small cavity, an ejection fraction of 70%, and deep narrow septal Q waves give hypertrophic cardiomyopathy, settled by high voltage with a thick wall. The loop shifts left with a steep end-diastolic curve, and the expected extra sound is an S4.

· Patients C and D have opposite voltages because patient C's thickness is extracellular amyloid, which adds mass without electrically active cells while myocytes atrophy, whereas patient D's thickness is muscle, which adds electrical mass and raises voltage.

· The ordered approach is to decide first whether emptying or filling fails, then to use wall thickness and the electrocardiogram to name the cause.

Distinctions that matter

· Ejection fraction separates systolic from diastolic failure; wall thickness with voltage then separates hypertrophic from restrictive disease.

· Asymmetry of thickening and systolic anterior motion are what mark hypertrophic disease, rather than thickness alone.

Common misconceptions

· Naming the disease from wall thickness first. Thickness is shared by hypertrophic and infiltrative disease, so the cycle half and the voltage are what decide.

· Reading an ejection fraction of 55% as excluding significant disease. In restrictive disease emptying is preserved and filling is the problem.

Where it is taught: Section C04, the comparison table and the worked example on classifying echocardiograms of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C04-CP02

Comparison table: The three cardiomyopathies side by side

A competent response contains

· Cavity size: enlarged in all four chambers in dilated disease; small left ventricular cavity in hypertrophic disease; normal or small with greatly enlarged atria in restrictive disease.

· Wall thickness: normal or thin in dilated disease; thick and often asymmetric at the septum in hypertrophic disease; normal or thick in restrictive disease, and thick in amyloid.

· Pattern of hypertrophy: eccentric with series sarcomeres in dilated disease; disarrayed and independent of load in hypertrophic disease; none in restrictive disease, where infiltration or fibrosis does the stiffening.

· Ejection fraction: reduced, usually under about 40%, in dilated disease; normal or high in hypertrophic disease; normal or near normal in restrictive disease.

· Primary dysfunction: systolic in dilated disease; diastolic in hypertrophic disease, with outflow obstruction in many; diastolic in restrictive disease.

· Pressure-volume loop: shifted right and narrow with a flatter end-systolic relationship in dilated disease; shifted left with a steep end-diastolic relationship in hypertrophic disease; a small loop with a steep end-diastolic relationship in restrictive disease.

· Extra heart sound: S3 in dilated disease; S4 in hypertrophic disease; S3 from the abrupt halt to filling in restrictive disease, with an S4 if the atria still contract.

· Electrocardiogram: nonspecific in dilated disease with arrhythmias and bundle branch block; high left ventricular voltage with septal Q waves in hypertrophic disease; low voltage in amyloid.

Distinctions that matter

· The end-diastolic pressure-volume relationship reports compliance and the end-systolic relationship reports contractility, so the two diastolic diseases steepen the first while the dilated ventricle flattens the second.

· Restrictive disease is the one in which a thick wall can coexist with low voltage, because the thickness is not muscle.

Common misconceptions

· Pairing a thick wall with a reduced ejection fraction. Thick-walled disease here preserves emptying and fails at filling.

· Assuming heart sounds alone separate the three. The S3 of dilated disease and the S3 of restrictive disease arise from different mechanisms, so they need the rest of the row to interpret.

Where it is taught: Section C04, the table of the three cardiomyopathies side by side and its caption of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-BD1

Brain dump 1 · foundational retrieval: Rebuild the causes and histology of myocarditis

A competent response contains

· Myocarditis is inflammation of the myocardium with injury or necrosis of myocytes that is not ischemic.

· Most cases in developed countries are viral: coxsackie B virus is the classic cause, with adenovirus, parvovirus B19, human herpesvirus 6, influenza, and SARS-CoV-2 among the others.

· Coxsackie B enters myocytes through the coxsackie-adenovirus receptor, which adenovirus also uses, and enteroviral protease 2A cleaves dystrophin, weakening the myocyte cytoskeleton.

· Myocytes are injured first by the virus and then by the cytotoxic T-cell response that clears it.

· Coxsackie B and other viruses: direct lysis plus T-cell injury, a lymphocytic infiltrate, often after a viral illness.

· Chagas disease from Trypanosoma cruzi: triatomine bug vector in Latin America, amastigotes in myocytes acutely, and chronic dilated cardiomyopathy with conduction block and an apical aneurysm years later, plus megaesophagus and megacolon.

· Lyme disease from Borrelia burgdorferi: weeks to months after the tick bite, with atrioventricular block of varying degree that usually resolves.

· Diphtheria from Corynebacterium diphtheriae: exotoxin ADP-ribosylates elongation factor 2, halting protein synthesis, with myocyte injury and conduction block.

· Drug hypersensitivity: interstitial infiltrate rich in eosinophils, often with rash and fever.

· Systemic lupus erythematosus: immune complex and autoantibody injury, which may coexist with pericarditis and Libman-Sacks endocarditis.

· Giant cell myocarditis: multinucleated giant cells with widespread necrosis in young to middle-aged adults, linked to autoimmunity and thymoma, and often fulminant.

· The defining histology is an inflammatory infiltrate with myocyte necrosis, which are the Dallas criteria.

· Viral infiltrate is mainly lymphocytes, patchy and interstitial, surrounding damaged myocytes; hypersensitivity disease is perivascular and interstitial with many eosinophils; giant cell disease shows multinucleated giant cells through extensive necrosis without the well-formed granulomas of sarcoidosis; diphtheria injures myocytes with relatively little inflammation.

· Because inflammation is often focal, a biopsy can miss it.

Distinctions that matter

· Eosinophils point to hypersensitivity, giant cells without well-formed granulomas point to giant cell disease, and parasites inside myocytes point to Chagas.

· Giant cell myocarditis and cardiac sarcoidosis both show giant cells, and the well-formed granulomas of sarcoidosis are what separate them.

Common misconceptions

· Treating myocarditis as purely a direct viral killing process. The cytotoxic T-cell response that clears the virus also destroys myocytes.

· Reading a negative biopsy as excluding the diagnosis. Involvement is patchy, so sampling can miss the inflamed tissue.

Where it is taught: Section C05, first paragraph, the causes table, and the histology paragraph of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-BD2

Brain dump 2 · integration and application: Explain the findings of myocarditis and its late outcome

A competent response contains

· Troponin rises because myocyte necrosis releases it, and here the necrosis is inflammatory rather than from coronary occlusion.

· Edema and patchy inflammation slow and block conduction and create foci of abnormal automaticity and reentry, which give palpitations, atrioventricular block, ventricular arrhythmias, and occasionally sudden death in a young person.

· The ejection fraction falls because widespread loss of contractile myocytes removes contractile tissue, which can reach fulminant cardiogenic shock.

· Diffuse ST elevation fits widespread myocardial involvement rather than a single coronary territory.

· Chest pain may be ischemic-like or pleuritic; a pleuritic character implies pericardial involvement as well, which is myopericarditis.

· Sequence to dilated cardiomyopathy: the virus enters myocytes through the coxsackie-adenovirus receptor; viral replication injures myocytes directly; cytotoxic T cells kill infected myocytes; exposed cardiac proteins can trigger persistent autoimmunity; lost myocytes are replaced by fibrosis; remodeling produces dilated cardiomyopathy.

· Replacement of lost myocytes by fibrosis is the step that makes the damage permanent, because scar cannot contract.

· Many eosinophils with rash and fever: drug hypersensitivity myocarditis, named by the eosinophil-rich infiltrate with the systemic allergic features.

· Multinucleated giant cells in widespread necrosis: giant cell myocarditis, named by giant cells without well-formed granulomas, often fulminant and linked to autoimmunity and thymoma.

· Parasites inside myocytes: Chagas disease, named by amastigotes of Trypanosoma cruzi in myocytes with exposure in Latin America.

Distinctions that matter

· A raised troponin reports myocyte death, not its mechanism, so inflammation and infarction can look the same on that one result.

· Myocarditis and myopericarditis differ in which layer is inflamed, and the character of the pain is the clue at the bedside.

Common misconceptions

· Reading ST elevation with a raised troponin as necessarily infarction. Diffuse changes with normal coronary arteries point elsewhere.

· Assuming recovery is impossible once the ejection fraction falls. Many patients recover fully, and it is the fibrotic replacement of myocytes that makes the loss permanent.

Where it is taught: Section C05, paragraph on manifestations, the flow from viral infection to dilated cardiomyopathy, and the causes table of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-RT02

Retrieval exercise: Rebuild the path from viral infection to dilated cardiomyopathy

A competent response contains

· Step 1: the virus enters myocytes through the coxsackie-adenovirus receptor, which coxsackie B virus and adenovirus both use.

· Step 2: viral replication injures myocytes directly, and enteroviral protease 2A cleaves dystrophin, weakening the cytoskeleton.

· Step 3: cytotoxic T cells kill infected myocytes while clearing the virus.

· Step 4: cardiac proteins exposed by that injury can trigger persistent autoimmunity, so damage continues after the virus has gone.

· Step 5: lost myocytes are replaced by fibrosis.

· Step 6: remodeling produces dilated cardiomyopathy.

· Fibrous tissue cannot contract, which is the mechanical property it lacks.

· The ejection fraction falls because contractile myocytes have been replaced by noncontractile scar, so the ventricle generates less force, empties less completely, and dilates.

Distinctions that matter

· Viral injury and the immune response are two separate routes to myocyte loss, and both operate in the same illness.

· Autoimmunity explains progression after viral clearance, which is why a resolved infection can still be followed by a failing ventricle.

Common misconceptions

· Stopping the sequence at viral clearance. Autoimmunity and fibrosis are what carry the process on to dilated disease.

· Treating the fibrosis as a repair that restores function. Scar fills the space without contributing contraction.

Where it is taught: Section C05, the flow from viral infection to dilated cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C05-CA01

Clinical application: A young man with chest pain, a raised troponin, and clean coronary arteries

A competent response contains

· Recognize: acute myocarditis, most likely viral given the preceding febrile illness with sore throat and diarrhea, and the most classic organism is coxsackie B virus, an enterovirus.

· Explain: inflammatory myocyte necrosis releases troponin without coronary occlusion; widespread loss and dysfunction of contractile myocytes give diffusely reduced wall motion and an ejection fraction of 34%;

edema and patchy inflammation slow and block conduction, which produces the intermittent atrioventricular delay; and the same inflamed tissue creates foci of abnormal automaticity and reentry, which produce the wide early beats.

· Differentiate: infarction would give regional wall motion loss in one coronary territory with obstructive disease on angiography, whereas the changes here are diffuse with normal coronary arteries; stress cardiomyopathy would show apical ballooning with a hypercontracting base after a catecholamine surge in an older, usually postmenopausal woman, with a troponin rise small relative to the dysfunction and no lymphocytic infiltrate; giant cell myocarditis would show multinucleated giant cells in widespread necrosis and hypersensitivity myocarditis would show an eosinophil-rich infiltrate, and the biopsy here shows neither, only lymphocytes.

· Investigate: the two features to name are an inflammatory infiltrate and myocyte injury or necrosis, which together are the Dallas criteria; a biopsy without them would not exclude myocarditis because involvement is often focal and sampling can miss the inflamed tissue.

· Decide: it can progress over months to years to dilated cardiomyopathy, with a large four-chamber dilated ventricle, thin or normal walls, a low ejection fraction, an S3, and functional mitral and tricuspid regurgitation.

· Justify: the virus enters myocytes through the coxsackie-adenovirus receptor, viral replication injures them directly, cytotoxic T cells kill infected myocytes, exposed cardiac proteins can trigger persistent autoimmunity, and lost myocytes are replaced by fibrosis that cannot contract; once contractility falls and the cavity enlarges, the law of Laplace means a larger radius with a thinner wall raises wall stress and oxygen demand, which injures more myocytes and drives further dilation.

Distinctions that matter

· Diffuse against regional dysfunction is what separates myocarditis from infarction, and the angiogram confirms it.

· The type of infiltrate names the kind of myocarditis, so lymphocytes, eosinophils, and giant cells point in three different directions.

Common misconceptions

· Taking a markedly raised troponin with ST elevation as proof of coronary occlusion in a young patient.

· Assuming progression to dilated disease is inevitable. Many patients recover fully, and progression depends on how many myocytes are lost and replaced by scar.

Where it is taught: Section C05, all paragraphs and the flow diagram, with the Laplace reasoning in Section C01 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild stress and tachycardia-induced cardiomyopathy

A competent response contains

· Stress cardiomyopathy is also called takotsubo cardiomyopathy and broken heart syndrome.

· About 90% of patients are women, most of them postmenopausal, and triggers are sudden emotional or physical stress such as bereavement, an argument, surgery, or severe illness.

· The typical pattern is apical ballooning: the apex and mid ventricle are akinetic while the base contracts vigorously, so in systole the ventricle resembles a narrow-necked Japanese octopus pot, the takotsubo.

· Mechanism: acute stress releases epinephrine and norepinephrine; very high epinephrine acts on dense apical beta receptors; beta-2 signaling switches from Gs to Gi and apical contraction falls; cAMP-driven calcium overload and microvascular spasm stun myocytes; the apex balloons while the base hypercontracts; function recovers over days to weeks.

· Tachycardia-induced cardiomyopathy is ventricular dysfunction caused by a persistently fast heart rate, most often atrial fibrillation or flutter with a rapid ventricular response, incessant supraventricular tachycardia, or very frequent premature ventricular contractions.

· A sustained high rate shortens diastole, which is the period of coronary perfusion and of calcium reuptake, while energy demand rises.

· The four myocardial changes are myocardial energy depletion, abnormal calcium handling, beta receptor downregulation, and subendocardial ischemia, which lead to dilation and a reduced ejection fraction resembling dilated cardiomyopathy.

· The shared feature that sets both conditions apart from the three classic cardiomyopathies is reversibility: takotsubo wall motion usually normalizes within weeks, and tachycardia-induced dysfunction usually recovers over weeks to months once the rate normalizes.

· The takotsubo mechanism is not fully settled, and catecholamine excess is the shared feature of all proposed explanations.

Distinctions that matter

· In takotsubo the myocytes are stunned rather than dead, which is why function returns; in dilated cardiomyopathy myocytes have been lost and replaced by scar.

· Rate itself can be the cause of a low ejection fraction, rather than only a consequence of a weak ventricle.

Common misconceptions

· Explaining apical ballooning by coronary occlusion. The akinetic region extends beyond any single coronary territory and the arteries are not obstructed.

· Treating a low ejection fraction in rapid atrial fibrillation as fixed muscle disease. Function often recovers once the rate is controlled.

Where it is taught: Section C06, first paragraph, the takotsubo mechanism flow, and the paragraph on tachycardia-induced cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-BD2

Brain dump 2 · integration and application: Separate a stunned ventricle from an infarcted one

A competent response contains

· Against infarction: the coronary arteries are normal, the akinetic region extends beyond the territory of any single coronary artery, the troponin rise is small relative to the large area of dysfunction, and the whole picture follows an acute emotional trigger with a long QT interval and deep T wave inversion.

· Biopsy shows contraction band necrosis, which is the pattern of catecholamine excess rather than of coronary occlusion.

· In six weeks her wall motion should be near normal, because catecholamine-stunned myocytes recover; they were not killed.

· An infarct of the same size would have left a permanent scar with persistent regional akinesis.

· Complications: heart failure and cardiogenic shock from the acutely weak apex and mid ventricle; dynamic outflow obstruction from the hypercontracting base narrowing the outflow tract; apical thrombus forming where blood stagnates against an akinetic apex; and ventricular arrhythmias, favored by the acute injury and the prolonged QT interval.

· The second patient has tachycardia-induced cardiomyopathy: an ejection fraction that recovers from 30% to 55% after rate control shows that the fast rate was the cause rather than primary muscle disease.

· Rate alone lowers the ejection fraction because a sustained high rate shortens diastole, which is the period of coronary perfusion and calcium reuptake, while raising energy demand, producing energy depletion, abnormal calcium handling, beta receptor downregulation, and subendocardial ischemia, with dilation and a fall in ejection fraction.

· Takotsubo mimics acute myocardial infarction at presentation with chest pain, dyspnea, ST elevation or deep T wave inversion, and a raised troponin.

Distinctions that matter

· Stunned myocardium recovers and infarcted myocardium scars, so the same initial tracing can have two very different outcomes.

· A troponin rise disproportionately small for the area of dysfunction is a useful pointer away from infarction.

Common misconceptions

· Expecting permanent dysfunction after takotsubo. Wall motion usually normalizes within weeks.

· Assuming a reduced ejection fraction during a sustained tachycardia means the arrhythmia was caused by a weak ventricle. The sequence can run the other way, and recovery after rate control shows which it was.

Where it is taught: Section C06, paragraph on takotsubo mimicking infarction, the stop and predict box, and the paragraph on tachycardia-induced cardiomyopathy of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C02-Q01

Board-style question: A systolic murmur that changes with position

Correct answer: C. Reduced left ventricular volume brings the anterior mitral leaflet closer to the hypertrophied septum

Why it is right: Standing lowers venous return and preload, so the left ventricle is smaller at the start of systole; a smaller cavity places the anterior mitral leaflet closer to the thickened septum, systolic anterior motion begins earlier, and the dynamic obstruction and its murmur both increase.

A is wrong: There is no fixed valvular narrowing in this patient, and in aortic stenosis standing reduces the volume crossing the orifice and makes the murmur softer rather than louder.

B is wrong: Standing reduces venous return rather than increasing it, and a larger end-diastolic volume would separate the leaflet from the septum and soften the murmur.

D is wrong: Contractility is not reduced on standing; sympathetic drive rises, and in this disease greater contractility worsens obstruction by emptying the ventricle further rather than by lengthening ejection.

A competent response contains

· Standing from squatting drops venous return, so left ventricular volume falls and the anterior mitral leaflet sits closer to the hypertrophied septum, worsening dynamic obstruction.

· Asymmetric septal thickening of 22 mm with a small cavity, a high ejection fraction, an S4, a murmur that does not radiate to the carotids, and a family history of sudden death all fit hypertrophic cardiomyopathy.

Distinctions that matter

· The obstruction is dynamic and depends on leaflet position, not on a fixed valve orifice.

Common misconceptions

· Treating every systolic murmur that changes with maneuvers as valvular. Here the obstruction is in the outflow tract and moves with cavity size.

Where it is taught: Section C02 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C03-Q02

Board-style question: A thick wall with small complexes

Correct answer: A. Extracellular protein deposits add wall thickness without adding electrically active myocytes

Why it is right: Extracellular amyloid increases measured wall thickness while adding no depolarizing cells, and the myocytes that remain atrophy and are separated by protein, so less electrically active muscle is available and QRS voltage falls despite the thick wall.

B is wrong: Myocyte disarray belongs to hypertrophic cardiomyopathy, which shows high left ventricular voltage with septal Q waves rather than low voltage, and the thickening there is muscle.

C is wrong: Series addition of sarcomeres is eccentric hypertrophy, which enlarges the cavity and leaves a normal or thin wall with a reduced ejection fraction, not a thick wall with a preserved one.

D is wrong: The pericardium is not the problem in this patient; pericardial disease limits filling without producing a thick, speckled myocardial wall, and the thick wall here is the finding that needs explaining.

A competent response contains

· A thick, speckled wall with low limb lead voltage, huge atria, a preserved ejection fraction, and right-sided congestion with the Kussmaul sign fit cardiac amyloidosis.

· Amyloid fibrils deposit between myocytes, adding mass without electrically active cells, while myocytes atrophy, so depolarizing muscle falls and QRS voltage drops.

Distinctions that matter

· Wall thickness on imaging reports anatomy; QRS voltage reports how much electrically active muscle is present.

Common misconceptions

· Assuming any thick-walled ventricle must show high voltage. That holds for true hypertrophy and is reversed by infiltration.

Where it is taught: Section C03 of the CV-18-E Primary Myocardial Diseases notes.

NPLEX-P1-CARD-18-05-C06-Q03

Board-style question: Akinesis beyond one coronary territory

Correct answer: D. Contraction band necrosis of myocytes without occlusive vascular disease

Why it is right: Stress cardiomyopathy is driven by a catecholamine surge, and biopsy in this condition shows contraction band necrosis, the histological signature of catecholamine excess, with no obstructive coronary disease to account for an akinetic region that extends beyond a single coronary territory.

A is wrong: An eosinophil-rich interstitial infiltrate indicates hypersensitivity myocarditis, which follows a drug exposure and usually comes with rash and fever rather than an acute emotional trigger and apical ballooning.

B is wrong: Congo red positive material with apple-green birefringence indicates cardiac amyloidosis, which presents with a thick wall, low QRS voltage, huge atria, and chronic right-sided congestion rather than acute akinesis with recovery.

C is wrong: Coagulative necrosis with neutrophils in a wedge-shaped region is the histology of myocardial infarction, which requires occlusive coronary disease and a region confined to one coronary territory, neither of which this patient has.

A competent response contains

· An acute emotional trigger, apical and mid ventricular akinesis with a hypercontracting base, a mildly raised troponin, a long corrected QT interval, and normal coronary arteries in a postmenopausal woman fit stress cardiomyopathy.

· The tissue pattern of catecholamine excess is contraction band necrosis, and there is no occlusive vascular disease to explain the dysfunction.

Distinctions that matter

· Contraction band necrosis reflects calcium overload from catecholamine excess, which is distinct from the coagulative necrosis of ischemic infarction.

Common misconceptions

· Expecting infarct histology because the tracing shows ST elevation. The arteries are open and the akinetic area crosses coronary territories.

Where it is taught: Section C06 of the CV-18-E Primary Myocardial Diseases notes.